

BUTTE MOONEY, MT, USAWMO: **727740**Lat: **45.965N**Lon: **112.501W**Elev: **5506**StdP: **12.00**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **24135****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.1	-8.7	-20.7	2.2	-14.9	-13.8	3.2	-8.1	22.7	32.1	19.7	31.3	2.7	160	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.1	88.1	57.4	85.0	56.5	82.0	55.9	60.3	79.6	58.9	78.1	57.4	76.4	10.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.5	77.7	62.7	52.1	71.1	62.2	49.9	65.4	61.8	29.0	79.3	27.9	78.0	26.9	76.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.2	16.1	DB	-27.4	92.5	7.5	2.6	-32.8	94.4	-37.2	95.9	-41.4	97.4	-46.9	99.3	(4)
(5)			WB	-25.9	63.9	5.9	4.6	-30.1	67.2	-33.6	69.8	-36.9	72.4	-41.2	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	20.4	22.1	31.5	38.7	47.3	55.6	64.3	61.9	52.8	40.8	28.7	19.0	(6)
(7)		DBStd	17.87	11.88	11.58	9.40	7.51	7.29	6.99	5.13	5.12	7.61	8.41	11.10	11.59	(7)
(8)		HDD50	4737	916	780	572	345	141	27	0	2	57	299	638	960	(8)
(9)		HDD65	9071	1381	1200	1037	788	549	290	73	118	370	752	1088	1425	(9)
(10)		CDD50	1225	0	0	0	6	57	194	444	371	140	13	0	0	(10)
(11)		CDD65	84	0	0	0	0	0	8	52	22	2	0	0	0	(11)
(12)		CDH74	3423	0	0	0	0	46	343	1622	1121	281	10	0	0	(12)
(13)		CDH80	1075	0	0	0	0	4	86	585	341	58	1	0	0	(13)
(14)	Wind	WSAvg	5.8	4.5	5.1	6.3	7.1	7.0	6.7	6.2	6.0	5.7	5.7	4.9	4.6	(14)
(15)	Precipitation	PrecAvg	12.2	0.6	0.4	0.7	1.0	1.9	2.2	1.3	1.4	1.1	0.7	0.5	0.4	(15)
(16)		PrecMax	20.0	1.7	0.9	1.9	2.6	4.8	5.8	4.2	3.1	3.1	3.1	1.3	1.1	(16)
(17)		PrecMin	8.1	0.0	0.0	0.2	0.1	0.5	0.3	0.1	0.2	0.1	0.0	0.2	0.1	(17)
(18)		PrecStd	3.1	0.4	0.2	0.4	0.6	0.9	1.3	1.0	0.8	0.8	0.6	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	51.0	62.7	71.2	79.2	87.6	93.0	90.3	86.0	74.8	61.2	47.1	(19)
(20)			MCWB	38.2	38.5	43.2	48.3	52.8	57.1	59.9	58.3	55.5	50.6	43.8	38.1	(20)
(21)		2%	DB	43.3	45.4	55.5	65.7	74.0	82.2	89.3	86.8	81.1	68.2	54.9	42.7	(21)
(22)			MCWB	35.5	36.4	40.7	45.7	51.2	55.8	58.3	56.8	54.4	47.7	41.3	35.7	(22)
(23)		5%	DB	40.1	41.9	51.2	60.3	69.9	78.0	86.2	84.0	77.1	62.8	49.3	39.2	(23)
(24)			MCWB	33.5	34.6	38.9	43.6	49.6	54.7	57.2	56.0	52.7	46.0	39.6	33.4	(24)
(25)		10%	DB	36.8	38.7	46.4	54.6	64.7	73.3	83.1	80.9	72.6	57.1	44.8	36.1	(25)
(26)			MCWB	31.3	32.3	36.6	41.1	47.9	52.9	56.4	55.5	51.3	43.7	36.8	31.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	40.0	44.3	49.6	55.7	59.7	62.8	61.8	59.4	51.8	46.0	39.0	(27)
(28)			MCD B	46.7	48.6	58.3	67.8	71.4	79.7	83.6	81.2	73.9	70.3	59.5	45.4	(28)
(29)		2%	WB	36.2	37.2	41.6	46.7	52.9	57.4	60.7	59.8	56.4	49.3	42.3	36.4	(29)
(30)			MCD B	42.2	44.3	54.3	62.9	69.2	76.2	80.8	78.4	74.0	64.9	52.4	42.0	(30)
(31)		5%	WB	33.9	35.0	39.4	44.3	50.9	55.8	59.3	58.3	54.5	47.0	39.9	33.8	(31)
(32)			MCD B	39.4	41.4	49.7	58.6	66.3	73.5	79.4	76.7	72.5	60.6	49.0	38.9	(32)
(33)		10%	WB	31.8	32.7	37.1	41.9	49.0	54.2	58.0	56.8	52.5	44.4	37.2	31.3	(33)
(34)			MCD B	36.7	38.2	45.8	53.6	63.3	70.5	77.9	75.3	69.9	56.1	44.3	35.8	(34)
(35)	Mean Daily Temperature Range		MDBR	21.7	22.8	22.8	23.3	24.9	27.1	34.1	34.0	30.4	25.6	23.0	21.1	(35)
(36)		5% DB	MCD BR	22.1	24.0	29.1	35.1	35.9	36.9	39.7	40.5	39.9	37.2	28.9	19.6	(36)
(37)			MCWBR	15.8	16.2	16.3	17.4	16.1	14.2	14.4	15.8	17.0	19.5	17.5	14.3	(37)
(38)			MCD BR	20.4	22.3	26.8	32.0	31.1	32.3	34.0	34.4	34.5	33.3	26.8	18.7	(38)
(39)		5% WB	MCWBR	15.0	15.2	15.6	16.5	14.7	13.1	13.0	13.7	14.7	17.8	16.6	13.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.235	0.244	0.259	0.286	0.294	0.304	0.310	0.319	0.290	0.272	0.246	0.233	(40)	
(41)		taud	2.524	2.529	2.588	2.512	2.559	2.564	2.559	2.496	2.571	2.604	2.596	2.550	(41)	
(42)		Ebn at Noon	289	306	314	310	309	305	301	295	298	290	282	278	(42)	
(43)		Edh at Noon	19	24	26	31	31	31	31	31	27	22	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	485	788	1190	1564	1803	2082	2273	1937	1445	911	538	387	(44)	
(45)		RadStd	44	63	69	107	158	160	128	102	128	73	46	28	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.04	-2.09	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	-0.58	-1.15	N/A	N/A	N/A	N/A	

Nomenclature: See separate page